Imperial College London

Department of Earth Science and Engineering

MSc in Geo-Energy with Machine Learning and Data Science

Independent Research Project

Project Plan

Developing Machine Learning Methods for Crystal Texture Analysis to Investigate Magmatic Evolution and the Onset of Explosive Volcanic Eruptions

by

Alexander Robert John Stringer

Email: alexander.stringer23@ic.ac.uk

GitHub Username: gems-ars223

GitHub Repository: https://github.com/ese-ada-lovelace-2025/irp-ars223.git

Supervisors:

Dr. Chetan Nathwani (Primary), Dr. Valentin Laurent (Secondary)

June 2026

Abstract

Plagioclase crystal textures are useful indicators of changing magmatic processes across geological time, offering insights into variations in eruptive behaviour useful for volcanic hazard assessment. Until now, studying such textures involved time and labour intensive manual petrographic methods affected by human bias. To expedite and increase the accuracy of such studies this project develops machine learning methods for automating the analysis of volcanic plagioclase crystals. Building upon an existing workflow (PlagNet), crystal images from cold cathodoluminescence (CCL) scans are extracted via deep-learning-based segmentation. Self-supervised learning (SSL) methods are then applied to the large unlabelled crystal image dataset learn meaningful crystal-texture representations. Suitable representation learning frameworks such as SimSiam, DeepCluster, SwAV and DINOv2 will be compared and the best performing model(s) will be optimised further. Learned embeddings will be analysed with clustering and dimensionality reduction techniques to investigate the correspondence between distinct crystal populations and different magmatic processes. By utilising recent advances in SSL to bypass the restrictions of manually labelled geological datasets previously unachievable insights into textural relationships can be produced. The final workflow will be a scalable framework for the quantitative analysis of volcanic plagioclase crystals contributing to a broader understanding of the ability of plagioclase crystals to record magmatic evolution.

1. Introduction

1.1 - Problem Outline

Volcanic crystal textures preserve key information regarding magmatic processes and are used to understand magma storage conditions, mixing and ascent dynamics (Higgins, 2017; Moschini et al., 2023). Changes in crystal texture can be linked to processes triggering explosive eruptions (e.g. magma recharge, fluid exsolution). Plagioclase crystal morphology, zoning and population structure are linked to eruptive behaviour and so aid the hazard assessment of volcanoes (Haller, n.d.; Toth and Maclennan, 2024). Manual petrographic analysis limits the scale and speed of much research and human bias is an unavoidable aspect of it which reduces accuracy. Therefore, this area of study would greatly benefit from the automation of such petrographic studies making them more scalable, efficient, accurate and unbiased. The use of SSL models and clustering would also make it possible to discover new relationships between texture and magmatic processes and further proof of existing ones.

1.2 - Literature Review

Deep learning has been applied to automate crystal segmentation from thin sections. Convolutional neural networks have been developed using

petrographic datasets to extract crystals images and measurements of textural properties (Leichter et al., 2022; Hassan et al., 2024; Halverson et al., 2025; Morales et al., 2025).

SSL allows for strong feature representation models to be trained without human-labelled data. SSL works to exploit the similarities between augmented versions of the same input image (contrastive) or the differences between different images (non-contrastive). Such models include SimCLR (Chen et al., 2020), SimSiam (Chen and He, 2021), BYOL (Grill et al., 2020), DeepCluster (Caron et al., 2018) and SwAV (Caron et al., 2020).

More recent, transformer-based approaches, such as DINO (Caron et al., 2021), DINOv2 (Oquab et al., 2023), DINOv3 (Rodgers, 2025) and Masked Autoencoders (He et al., 2022) have achieved remarkable success and high levels of generalization across domains.

SSL is increasingly used in scientific image analysis where labelled datasets are limited. SSL applied in non-geoscience but image focused tasks in fields such as histopathology and cell microscopy shows how SSL may be used within this project. Studies show SSL can successfully reconstruct biologically relevant structure without manual annotation (Ciga et al., 2020; Galusser et al., 2023; Bendidi et al., 2024; Moutakanni et al., 2025; Rodgers, 2025). There are some similar studies in adjacent geoscience fields like palaeontology and mining geology too (Ferreira, 2023; Koeshidayatullah and Ferreira, 2023; Rodgers, 2025). SSL's success in similar domains suggests potential for transferability to CCL thin section analysis which faces similar issues regarding data acquisition, complexity, and lack of labels.

Within volcanology, machine learning has recently expanded significantly providing novel ways to analyse volcanic crystals and support interpretations of volcanic systems. For example, recent work demonstrates automated crystal population analysis and the measurement of their size distributions for rapid textural investigation (Jutzeler et al., 2024; Toth and Maclennan, 2024; Halverson et al., 2025); However, these methods are mostly supervised or semi-supervised so lack the capacity to efficiently learn from vast unlabelled datasets.

Overall, there is a substantial gap in the literature between cutting-edge SSL implementations (Chen et al., 2020; Chen and He, 2021; Caron et al., 2021; Oquab et al., 2023) and its application to volcanic crystal texture analysis (Toth and Maclennan, 2024; Morales et al., 2025). A successful implementation of SSL to a segmentation workflow such as PlagNet (Nathwani and Peng, 2026) offers a novel way to achieve the analysis of crystal textures at unprecedented scale and may provide the capability to explore previously undetectable patterns linked to eruptive behaviour. Finally, this development process can benefit greatly from key lessons learnt from the application of SSL models to similar image data in other fields (Ciga et al., 2020; Galusser et al., 2023; Ferreira, 2023; Koeshidayatullah and Ferreira, 2023; Bendidi et al., 2024; Moutakanni et al., 2025; Rodgers, 2025).

1.3 - Methodology

1. **Investigating Existing Preprocessing and Segmentation**

Preprocess and segment CCL thin section images producing plagioclase images using a deep-learning based segmentation model. Post-processing will remove artefacts and produce descriptors for crystal size and shape. Optimise PlagNet code to improve segmentation and therefore data quality.

2. **Development and Comparison of Self-Supervised Learning (SSL)**

Suitable SSL models (SimSiam, SwAV, DINOv2, DeepCluster) will be trained on segmented crystal images without manual labels and compared for performance. SimSiam SwAV and DeepCluster have been chosen for their ability to produce good results using a comparable amount of data (10,000-100,000 images) and because their compute requirements align with the compute available. DINOv2 will be used as a pretrained model.

3. **Clustering and Analysis of Embeddings**

Learned embeddings are analysed with unsupervised clustering methods (i.e. k-means) and dimensionality reduction (i.e. UMAP and PCA) to identify distinct crystal populations and allowing the visualisation of structure in the data.

4. **Optimisation of Chosen Model**

Once a strong model has been identified and initial results have been produced, the model will be optimised in order to refine results.

5. **Interpretation within Geological Context**

Any results suggesting relationships between crystal populations and magmatic processes will be interpreted using knowledge of the data sources (Aegina, Methana, Mt St Helens and Rumania).

2. Progress at the Time of This Report

Previously, a large dataset of CCL thin sections was collected and plagioclase images have been segmented (>12,000 images) from them by the creators of the PlagNet repository (Peng et al., 2026). PlagNet also provides a working pipeline for pre-processing and segmenting CCL thin-section images and a cursory attempt at training a SimSiam SSL model was completed. Therefore, the project from now will aim to optimise the current segmentation method, improve upon the existing PlagNet pipeline and develop and compare suitable SSL methods. PlagNet and all notebooks have been tested and confirmed to be in working order. Preliminary development of new SSL methods has begun. Model selection has been informed by extensive reading into SSL methods on which models are most suitable to trial for this task. The chosen models are SimSiam, DINOv2, DeepCluster and SwAV.

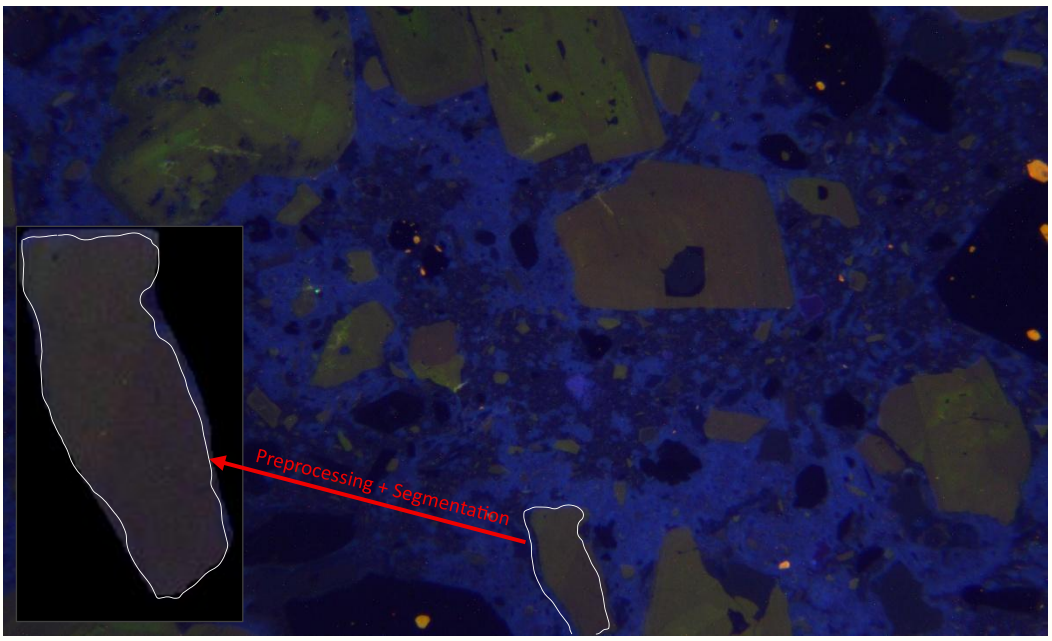


***Figure 1**: Example of raw CCL thin section image data (centre) and the segmented plagioclase image data (bottom left) which results from using the pre-existing PlagNet pipeline on the raw data.*

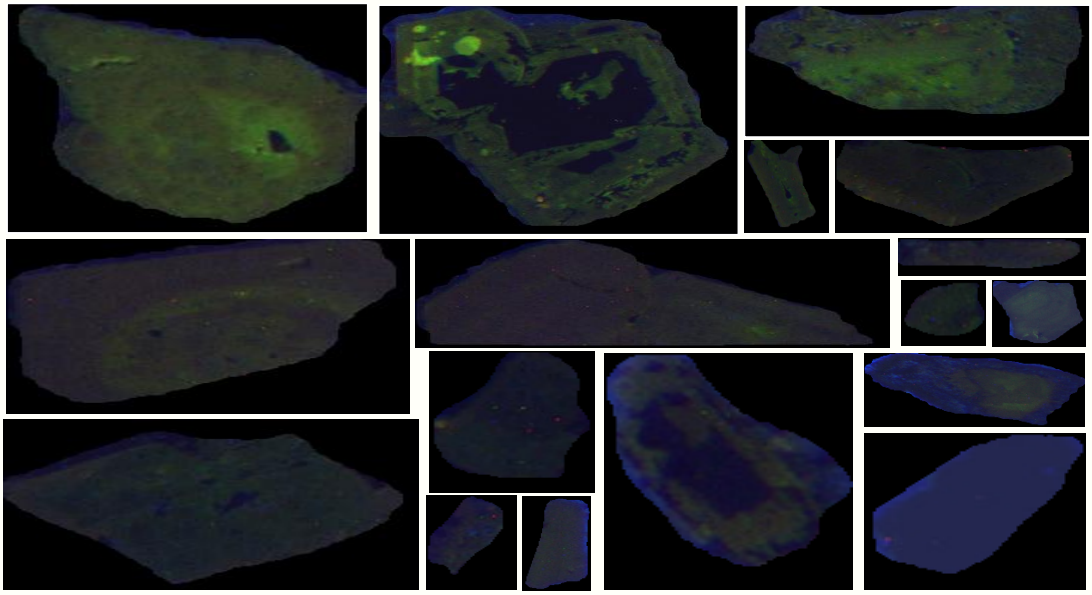


***Figure 2**: Examples of variance in size, shape, colour and texture of segmented plagioclase crystals to be sorted during clustering.*

Model	SSL Type	Compute Cost	Images Needed
SimSiam (Chen and He, 2021)	Siamese	Low-Medium	Moderate (~10,000-100,000)
SimCLR (Chen at al., 2020)	Contrastive	High	Large (50,000-1,000,000+)
DeepCluster (Caron et al., 2018)	Clustering Based	Medium	Moderate (~10,000-100,000)
SwAV (Caron et al., 2020)	Online Clustering	Medium-High	Moderate-Large (~10,000-500,000)
DINO (Caron et al., 2021)	Self-Distillation ViT (Vision Transformer)	High	Large (50,000-1,000,000+)
DINOv2 (Oquab et al., 2023)	Self-Distillation ViT (Vision Transformer)	High (pretrained)	Pretrained (on 142,000,000)
DINOv3 (Rodgers, 2025)	Self-Distillation ViT (Vision Transformer)	Very High	Extremely Large (100,000,000-1,000,000,000+)
BYOL (Grill et al., 2020)	Bootstrap	Moderate	Moderate-Large (~10,000-500,000)

***Figure 3**: Chart comparing SSL models considered for this project. The models chosen due to high suitability for the task are SimSiam, SwAV, DeepCluster and DINOv2.*

3. Objectives for the Rest of This Project

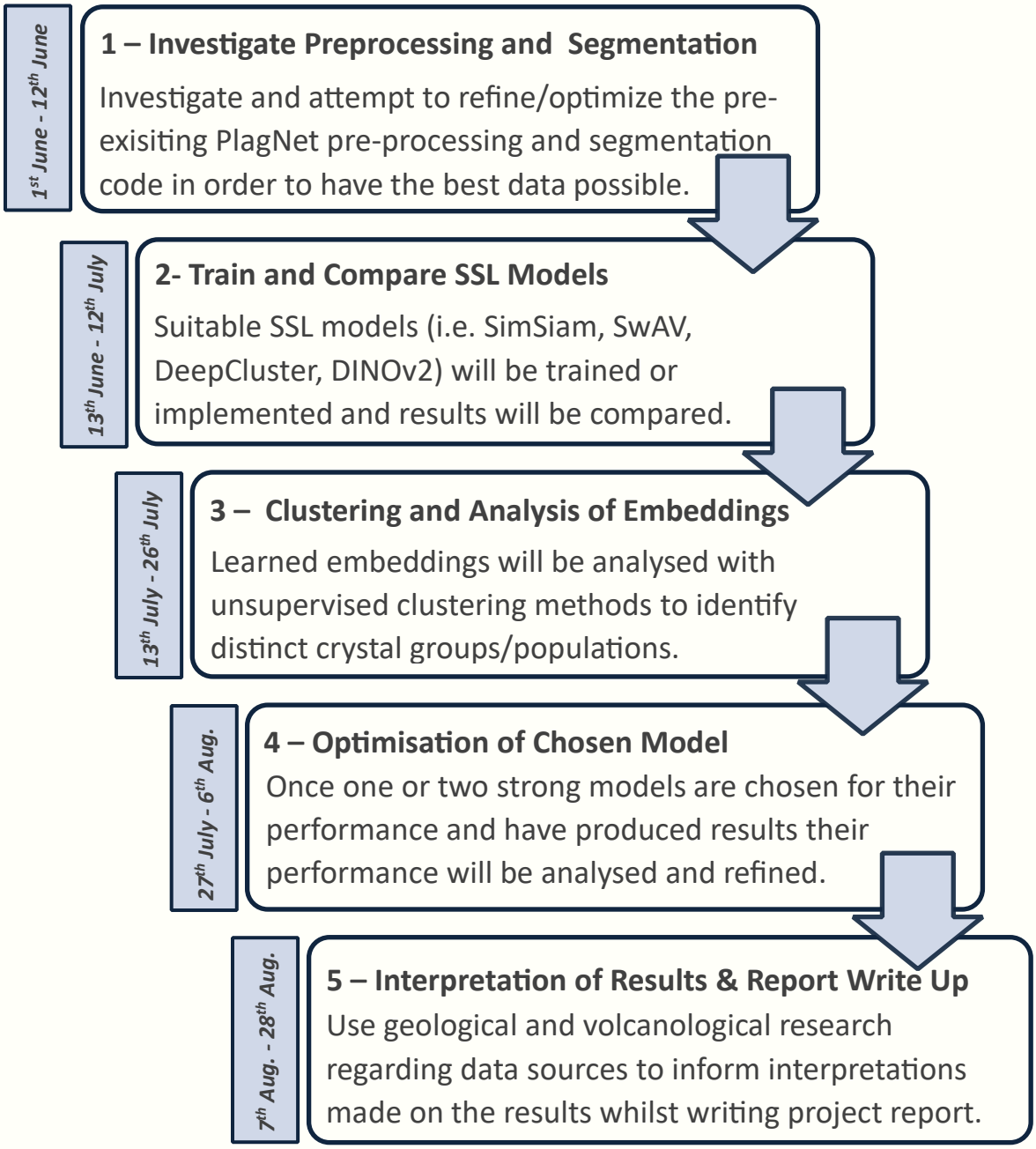


Figure 4: *Flow chart outlining major objectives left for this research project.*

References

Bendidi, I., Bardes, A., Cohen, E., Lamiable, A., Bollot, G. and Genovesio, A. (2024) 'Exploring self-supervised learning biases for microscopy image representation', *Biological Imaging*, 4, e12. doi:10.1017/S2633903X2400014X.

Caron, M., Bojanowski, P., Joulin, A. and Douze, M. (2018) 'Deep Clustering for Unsupervised Learning of Visual Features', in *Proceedings of the European Conference on Computer Vision (ECCV)*, pp. 132–149.

Caron, M., Misra, I., Mairal, J., Goyal, P., Bojanowski, P. and Joulin, A. (2020) 'Unsupervised Learning of Visual Features by Contrasting Cluster Assignments', *Advances in Neural Information Processing Systems*, 33, pp. 9912–9924.

Caron, M., Touvron, H., Misra, I., Jégou, H., Mairal, J., Bojanowski, P. and Joulin, A. (2021) 'Emerging Properties in Self-Supervised Vision Transformers', in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 9650–9660.

Chen, T., Kornblith, S., Norouzi, M. and Hinton, G. (2020) 'A Simple Framework for Contrastive Learning of Visual Representations', in *Proceedings of the 37th International Conference on Machine Learning (ICML)*, pp. 1597–1607.

Chen, X. and He, K. (2021) 'Exploring Simple Siamese Representation Learning', in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 15750–15758. doi:10.1109/CVPR46437.2021.01549.

Ciga, O., Xu, T. and Martel, A.L. (2020) 'Self-supervised contrastive learning for digital histopathology', *arXiv preprint*, arXiv:2011.13971.

Ferreira, I.E. (2023) 'No Labels Needed: Self-Supervised Semantic Segmentation of Ore Minerals in Thin Section', *SSRN Electronic Journal*. doi:10.2139/ssrn.4466357.

Gallusser, B., et al. (2023) ‘Self-supervised dense representation learning for live-cell microscopy image analysis with time-arrow prediction’, *arXiv preprint*, arXiv:2305.05511.

Grill, J.-B., Strub, F., Altché, F., Tallec, C., Richemond, P.H., Buchatskaya, E., Doersch, C., Pires, B., Guo, Z., Azar, M.G., Piot, B., Kavukcuoglu, K., Munos, R. and Valko, M. (2020) ‘Bootstrap Your Own Latent: A New Approach to Self-Supervised Learning’, *Advances in Neural Information Processing Systems*, 33, pp. 21271–21284.

Haller, L.L. (n.d.) *Investigating magmatic processes using cold-cathodoluminescence imaging of plagioclase*. Available at: https://tinyurl.com/4s8erysv (Accessed: 26 May 2026).

Halverson, B., Loewen, M., Dietterich, H. and Whittington, A. (2025) ‘Case study of deep learning image segmentation for the purposes of rapid 2D petrographic analysis in volcanic rocks’, *Volcanica*, 8(2), pp. 427–442. doi:10.30909/vol/gsfc1696.

Hassan, S.M., Laban, N., Abo Khashaba, S.M., El-Shibiny, N.H., Bashir, B., Azer, M.K., Drüppel, K. and Keshk, H.M. (2024) ‘Semantic Segmentation of Some Rock-Forming Mineral Thin Sections Using Deep Learning Algorithms: A Case Study from the Nikeiba Area, South Eastern Desert, Egypt’, *Remote Sensing*, 16(13), p. 2276. doi:10.3390/rs16132276.

He, K., Chen, X., Xie, S., Li, Y., Dollár, P. and Girshick, R. (2022) ‘Masked Autoencoders Are Scalable Vision Learners’, in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 16000–16009. doi:10.1109/CVPR52688.2022.01553.

Higgins, M.D. (2017) ‘Quantitative investigation of felsic rock textures using cathodoluminescence images and other techniques’, *Lithos*, 277, pp. 259–268. doi:10.1016/j.lithos.2016.05.006.

Jutzeler, M., Carey, R.J., Dagasan, Y., McNeill, A. and Cas, R.A.F. (2024) ‘Machine-learning crystal size distribution for volcanic stratigraphy correlation’, *Scientific Reports*, 14(1). doi:10.1038/s41598-024-82847-0.

Koeshidayatullah, A. and Ferreira, I. (2023) 'SelfMin: Self-Supervised Deep Learning for Advanced Mineralogical Analysis', *EGU General Assembly 2023*. doi:10.5194/egusphere-egu23-2189.

Leichter, A., Almeev, R.R., Wittich, D., Beckmann, P., Rottensteiner, F., Holtz, F. and Sester, M. (2022) 'Automated Segmentation of Olivine Phenocrysts in a Volcanic Rock Thin Section Using a Fully Convolutional Neural Network', *Frontiers in Earth Science*, 10. doi:10.3389/feart.2022.740638.

Morales, J., Saldivia, C., Lobo, R., del Pino, M., Muñoz, M., Caniggia, G., Catalán, J., Hayde, R. and Poblete, V. (2025) 'Visual analysis of deep learning semantic segmentation applied to petrographic thin sections', *Scientific Reports*, 15(1). doi:10.1038/s41598-025-99767-2.

Moschini, P., Mollo, S., Pontesilli, A., Nazzari, M., Petrone, C., Fanara, S., Vona, A., Gaeta, M., Romano, C. and Scarlato, P. (2023) 'A review of plagioclase growth rate and compositional evolution in mafic alkaline magmas: Guidelines for thermometry, hygrometry, and timescales of magma dynamics at Stromboli and Mt. Etna', *Earth-Science Reviews*, 240, p. 104399. doi:10.1016/j.earscirev.2023.104399.

Moutakanni, T., Couprie, C., Yi, S., Doron, M., Chen, Z.S., Moshkov, N., Gardes, E., Caron, M., Touvron, H., Joulin, A., Bojanowski, P., Pernice, W.M. and Caicedo, J.C. (2025) 'Cell-DINO: Self-supervised image-based embeddings for cell fluorescent microscopy', *PLoS Computational Biology*, 21(12), e1013828. doi:10.1371/journal.pcbi.1013828.

Peng, A., Nathwani, C., Rigaux, A. (2026) *PlagNet: Reading the Crystal Record of Volcanic Eruptions* [computer software]. GitHub. Available at: https://github.com/ChetanNathwani/PlagNet (Accessed: 20 May 2026).

Oquab, M., Darcet, T., Moutakanni, T., Vo, H., Szafraniec, M., Khalidov, V., Fernandez, P., Haziza, D., Massa, F., El-Nouby, A., Assran, M., Ballas, N., Galuba, W., Howes, R., Huang, P.-Y., Li, S.-W., Misra, I., Rabbat, M., Sharma, V., Synnaeve, G., Xu, H., Jégou, H., Mairal, J., Labatut, P., Joulin, A. and

Bojanowski, P. (2023) ‘DINOv2: Learning Robust Visual Features without Supervision’, *arXiv preprint*, arXiv:2304.07193.

Rodgers, N. (2025) ‘Complexity Begets Simplicity: Self-Supervised Learning for Palaeontological Images with Few or No Labels’, *bioRxiv*. doi:10.1101/2025.11.13.688022.

Toth, N. and Maclennan, J. (2024) ‘MinDet1: A deep learning-enabled approach for plagioclase textural studies’, *Volcanica*, 7(1), pp. 135–151. doi:10.30909/vol.07.01.135151.

4. AI Acknowledgement

Services: ChatGPT, Claude

Producers: OpenAI, Anthropic

Links: https://chatgpt.com/, https://claude.ai/

How generative AI was used: Used occasionally to understand jargon and notation from some papers which had unfamiliar technical language and to query issues with grammar or punctuation.

Declaration regarding AI use: All work in this project plan has been produced and written solely by the author listed above despite the use of the Generative AI tool listed above.